

STAT6018 Research Frontiers in Data Science

Topic I: Statistical methods for analyzing complex survival data

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Table of Contents

- 1 Chapter 2: Semiparametric regression analysis of interval-censored data
 - Transformation models for interval-censored data
 - Transformation models for multivariate interval-censored data
 - Marginal models for multivariate interval-censored data
 - Semiparametric regression models for panel count data

Table of Contents

- ① Chapter 2: Semiparametric regression analysis of interval-censored data
 - Transformation models for interval-censored data
 - Transformation models for multivariate interval-censored data
 - Marginal models for multivariate interval-censored data
 - Semiparametric regression models for panel count data

Reference



Zeng, D., Mao, L., & Lin, D. Y. (2016). Maximum likelihood estimation for semiparametric transformation models with interval-censored data. *Biometrika*, 103(2), 253-271.

Interval-censored data

- Interval-censored data arise when the failure time is only known to lie within a broad time interval.
- Commonly encountered when the disease onset can only be ascertained through a small number of examinations.
 - ▶ HIV infection: periodic blood tests
 - ▶ Alzheimer's disease onset: periodic cognitive tests
 - ▶ Tumor occurrence: biopsies at periodic clinical visits
- Types of interval-censored data:
 - ▶ Case 1: only one examination time per subject, aka **current status data**
 - ▶ Case k ($k \geq 2$): k examination times per subject¹
 - ▶ Mixed case: number of examination times varies among subjects
- Theoretical and computational challenges: no exact failure time

¹ Huang, J., & Wellner, J. A. (1997). Interval censored survival data: a review of recent progress. In *Proceedings of the first Seattle symposium in biostatistics: survival analysis* (pp. 123-169). New York, NY: Springer US.

Transformation models

Notation:

- T : failure time
- $X(t)$: potentially time-dependent covariates
- $\Lambda(t|X)$: conditional cumulative hazard function for T given $X(\cdot)$

Semiparametric transformation model:

$$\Lambda(t|X) = G \left[\int_0^t \exp \{ \beta^T X(s) \} d\Lambda(s) \right]$$

- $G(\cdot)$: strictly increasing transformation function
 - ▶ $G(x) = x \Rightarrow$ proportional hazards model
 - ▶ $G(x) = \log(1 + x) \Rightarrow$ proportional odds model
- β : unknown regression parameters
- $\Lambda(\cdot)$: unknown increasing function

Frailty-induced transformations

Log-Laplace transform:

$$G(x) = -\log \int_0^{\infty} e^{-x\xi} f(\xi) d\xi$$

- ξ : frailty variable with support $[0, \infty)$
- $f(\xi)$: density function of ξ
 - ▶ Gamma density with mean 1 and variance $r \Rightarrow$ logarithmic transformations $G(x) = r^{-1} \log(1 + rx)$ ($r \geq 0$)
 - ▶ Positive stable distribution with parameter $\rho < 1 \Rightarrow$ Box-Cox transformations $G(x) = \{(1 + x)^\rho - 1\}/\rho$

Data

Raw data:

- Examination times: $U = (0 = U_0, U_1, \dots, U_M, U_{M+1} = \infty)$
- Event statuses: $\Delta = (\Delta_0, \Delta_1, \dots, \Delta_M)$, with $\Delta_l = I(U_l < T \leq U_{l+1})$
- Covariates: $X(t)$

Independent censoring assumption: $(U, M) \perp\!\!\!\perp T$ conditional on $X(t)$

Simplified data: (L, R, X)

- $L = \max\{U_l : U_l < T\}$ and $R = \min\{U_l : U_l \geq T\}$
- $L = 0 \Rightarrow$ left censoring
- $R = \infty \Rightarrow$ right censoring

Data from n independent samples: $\{\mathcal{O}_i = (L_i, R_i, X_i) : i = 1, \dots, n\}$

Likelihood

Observed-data likelihood:

$$L_n(\beta, \Lambda) = \prod_{i=1}^n \left(\exp \left[-G \left\{ \int_0^{L_i} e^{\beta^\top X_i(s)} d\Lambda(s) \right\} \right] - \exp \left[-G \left\{ \int_0^{R_i} e^{\beta^\top X_i(s)} d\Lambda(s) \right\} \right] \right)$$

NPMLE:

$$\tilde{L}_n(\beta, \Lambda) = \prod_{i=1}^n \left[\exp \left\{ -G \left(\sum_{t_l \leq L_i} \lambda_l e^{\beta^\top X_{il}} \right) \right\} - I(R_i < \infty) \exp \left\{ -G \left(\sum_{t_l \leq R_i} \lambda_l e^{\beta^\top X_{il}} \right) \right\} \right]$$

- $t_1 < \dots < t_m$: distinct values of all $L_i > 0$ and $R_i < \infty$ ($i = 1, \dots, n$)
- λ_l : jump size of Λ at t_l ($l = 1, \dots, m$)
- $X_{il} = X_i(t_l)$

Poisson data augmentation

- Consider the class of frailty-induced transformations, $\tilde{L}_n(\beta, \Lambda)$ can be written as

$$\prod_{i=1}^n \int_{\xi_i} \underbrace{\exp\left(-\xi_i \sum_{t_l \leq L_i} \lambda_l e^{\beta^\top X_{il}}\right) \left\{ 1 - \exp\left(-\xi_i \sum_{L_i < t_l \leq R_i} \lambda_l e^{\beta^\top X_{il}}\right) \right\}^{I(R_i < \infty)}}_{p(\mathcal{O}_i | \xi_i)} f(\xi_i) d\xi_i$$

- Direct maximization of $\tilde{L}_n(\beta, \Lambda)$ over β and λ_l is difficult.
 - Lack of analytical expressions for λ_l
 - Many λ_l are zero and lie on the boundary of the parameter space
- We introduce latent variables $W_{il} \stackrel{\text{ind}}{\sim} \text{Poisson}(\xi_i \lambda_l e^{\beta^\top X_{il}})$. Then $p(\mathcal{O}_i | \xi_i)$ is equivalent to the probability of the event

$$\tilde{\mathcal{O}}_i = \left(\sum_{t_l \leq L_i} W_{il} = 0 \right) \cap \left(\sum_{L_i < t_l \leq R_i} W_{il} > 0 \right)^{I(R_i < \infty)}$$

EM algorithm

- Therefore, maximizing $\tilde{L}_n(\beta, \Lambda)$ is equivalent to maximizing the likelihood based on $\tilde{\mathcal{O}}_i$ ($i = 1, \dots, n$).
- The maximization can be solved via an EM algorithm, treating ξ_i and W_{il} as missing data.
- Define $R_i^* = L_i I(R_i = \infty) + R_i I(R_i < \infty)$. The complete-data log-likelihood is

$$\sum_{i=1}^n \left[\sum_{l=1}^m I(t_l \leq R_i^*) \left\{ W_{il} \log(\xi_i \lambda_l e^{\beta^\top x_{il}}) - \xi_i \lambda_l e^{\beta^\top x_{il}} - \log W_{il}! \right\} + \log f(\xi_i) \right]$$

E-step

- In the E-step, we evaluate the posterior means $\hat{E}(\xi_i)$ and $\hat{E}(W_{il})$.
- The posterior density of ξ_i is proportional to $p(\mathcal{O}_i|\xi_i)f(\xi_i)$. Simple algebra yields

$$\hat{E}(\xi_i) = \frac{\exp\{-G(S_{i1})\} G'(S_{i1}) - I(R_i < \infty) \exp\{-G(S_{i2})\} G'(S_{i2})}{\exp\{-G(S_{i1})\} - I(R_i < \infty) \exp\{-G(S_{i2})\}},$$

where $S_{i1} = \sum_{t_l \leq L_i} \lambda_l e^{\beta^\top x_{il}}$ and $S_{i2} = \sum_{t_l \leq R_i} \lambda_l e^{\beta^\top x_{il}}$.

E-step (cont.)

- Clearly, $\hat{E}(W_{il}) = 0$ if $t_l \leq L_i$.
- For $L_i < t_l \leq R_i$ with $R_i < \infty$,

$$\begin{aligned}\hat{E}(W_{il}) &= E_{\xi_i} \left\{ E(W_{il} | \tilde{\mathcal{O}}_i, \xi_i) \middle| \tilde{\mathcal{O}}_i \right\} \\ &= E_{\xi_i} \left\{ E \left(W_{il} \middle| \sum_{L_i < t_{l'} \leq R_i} W_{il'} > 0, \xi_i \right) \middle| \mathcal{O}_i \right\} \\ &= \hat{E} \left[\frac{\xi_i \lambda_l e^{\beta^T X_{il}}}{1 - \exp \{ -\xi_i (S_{i2} - S_{i1}) \}} \right]\end{aligned}$$

- The integral over ξ_i can be approximated by Gaussian–Laguerre quadrature.

M-step

- In the M-step, we first update λ_l by

$$\lambda_l = \frac{\sum_{i=1}^n I(t_l \leq R_i^*) \hat{E}(W_{il})}{\sum_{i=1}^n I(t_l \leq R_i^*) \hat{E}(\xi_i) e^{\beta^\top X_{il}}}, \quad \text{for } l = 1, \dots, m$$

- After plugging the above λ_l into the conditional expectation of the complete-data log-likelihood, we can then update β by solving the equation

$$\sum_{i=1}^n \sum_{l=1}^m I(t_l \leq R_i^*) \hat{E}(W_{il}) \left\{ X_{il} - \frac{\sum_{j=1}^n I(t_l \leq R_j^*) \hat{E}(\xi_j) e^{\beta^\top X_{jl}} X_{jl}}{\sum_{j=1}^n I(t_l \leq R_j^*) \hat{E}(\xi_j) e^{\beta^\top X_{jl}}} \right\},$$

which can be solved using the one-step Newton-Raphson method.

Remarks

- By introducing Poisson variables, we turn the original nonconcave likelihood function to a weighted sum of Poisson log-likelihood functions, which is **strictly concave**.
- In the M-step, the high-dimensional parameters λ_l ($l = 1, \dots, m$) have **closed-form solutions**. This avoids the inversion of any large Hessian matrices.
- The observed-data likelihood is guaranteed to increase after each iteration of the EM algorithm.

Variance estimation

- We use profile likelihood² to estimate the covariance matrix of $\hat{\beta}$.
- Define the profile log-likelihood as

$$\text{pl}_n(\beta) = \max_{\Lambda} \log L_n(\beta, \Lambda),$$

which can be computed using the same EM algorithm but with fixed β .

- The covariance matrix of $\hat{\beta}$ can be estimated by

$$\hat{V} = - \left[\left\{ \frac{\text{pl}_n(\hat{\beta}) - \text{pl}_n(\hat{\beta} + h_n e_j) - \text{pl}_n(\hat{\beta} + h_n e_k) + \text{pl}_n(\hat{\beta} + h_n e_j + h_n e_k)}{h_n^2} \right\}_{(j,k)} \right]^{-1},$$

where e_j is the j th canonical vector and h_n is a constant of order $n^{-1/2}$.

²Murphy, S. A., & Van der Vaart, A. W. (2000). On profile likelihood. *Journal of the American Statistical Association*, 95(450), 449-465.

Asymptotic theory

Consistency:

$$\|\hat{\beta} - \beta_0\| + \sup_{t \in [0, \tau]} |\hat{\Lambda}(t) - \Lambda_0(t)| \xrightarrow{a.s.} 0$$

Asymptotic normality & semiparametric efficiency:

$$\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} N(0, \tilde{\mathcal{I}}_0^{-1}),$$

where $\tilde{\mathcal{I}}_0$ is the efficient information matrix of β .

Consistency of variance estimator: $\|n\hat{V} - \tilde{\mathcal{I}}_0^{-1}\|_2 = o_p(1)$.

Mixed rate of convergence:

$$E \left[\sum_{l=1}^M \left\{ \int_0^{U_l} e^{\hat{\beta}^\top X(s)} d\hat{\Lambda}(s) - \int_0^{U_l} e^{\beta_0^\top X(s)} d\Lambda_0(s) \right\}^2 \right]^{1/2} = O_p(n^{-1/3})$$

Table of Contents

- 1 Chapter 2: Semiparametric regression analysis of interval-censored data
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 - Transformation models for multivariate interval-censored data
 - Marginal models for multivariate interval-censored data
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Reference

-  Zeng, D., Gao, F., & Lin, D. Y. (2017). Maximum likelihood estimation for semiparametric regression models with multivariate interval-censored data. *Biometrika*, 104(3), 505-525.

Motivation

- Interval-censored multiple-event data
 - ▶ Study of chronic diseases: diabetes, hypertension, Alzheimer's disease
- Interval-censored clustered data
 - ▶ Twin/family study of infectious diseases
 - ▶ Dental caries
- Challenges:
 - ▶ Event times never exactly observed
 - ▶ Dependence between events from the same subject
 - ▶ Dependence within clusters

Transformation models

Notation:

- n : number of independent clusters
- J_i : number of subjects in the i th cluster
- K : number of event types
- T_{ijk} : k th event time for the j th subject of the i th cluster
- $X_{ijk}(t)$: potentially time-dependent covariates
- $b_i \sim N(0, \Sigma_i(\gamma))$: vector of random effects

Semiparametric transformation model:

$$\Lambda_{ijk}(t|X_{ijk}, b_i) = G_k \left[\int_0^t \exp \{ \beta^\top X_{ijk}(s) + b_i^\top Z_{ijk}(s) \} d\Lambda_k(s) \right]$$

- $G_k(\cdot)$: type-specific transformation function
- β, γ : unknown regression parameters
- $Z_{ijk}(\cdot)$: contains 1 and part of $X_{ijk}(\cdot)$
- $\Lambda_k(\cdot)$: arbitrary increasing function

Transformation models (cont.)

Semiparametric transformation model:

$$\Lambda_{ijk}(t|X_{ijk}, b_i) = G_k \left[\int_0^t \exp \{ \beta^\top X_{ijk}(s) + b_i^\top Z_{ijk}(s) \} d\Lambda_k(s) \right] \quad (1)$$

- By letting X_{ijk} and Z_{ijk} depend on k , model (1) allows the regression parameters and random effects to vary across the K types of events.
- The dependence of Z_{ijk} on j allows for subject-specific random effects.
- $\Sigma_i(\gamma)$ usually does not depend on i , such that γ contains the upper diagonal elements of the common covariance matrix Σ .

Data

Examination times for T_{ijk} :

$$U_{ijk} = (0 = U_{ijk0}, U_{ijk1}, \dots, U_{ijk, M_{ijk}}, U_{ijk, M_{ijk}+1} = \infty)$$

Data:

$$\left\{ \mathcal{O}_{ijk} = (L_{ijk}, R_{ijk}, X_{ijk}) : i = 1, \dots, n; j = 1, \dots, J_i; k = 1, \dots, K \right\},$$

where $(L_{ijk}, R_{ijk}]$ is the shortest time interval induced by U_{ijk} that brackets T_{ijk} .

Independent censoring assumption:

$\{(U_{ijk}, M_{ijk}) : j = 1, \dots, J_i; k = 1, \dots, K\}$ are independent of
 $\{T_{ijk} : j = 1, \dots, J_i; k = 1, \dots, K\}$ and b_i
conditional on $\{X_{ijk}(\cdot) : j = 1, \dots, J_i; k = 1, \dots, K\}$.

Likelihood

Let $\theta = (\beta^\top, \gamma^\top)^\top$ and $\mathcal{A} = \{\Lambda_k\}_{k=1}^K$. The likelihood is

$$L_n(\theta, \mathcal{A}) = \prod_{i=1}^n \int_{b_i} \prod_{j=1}^{J_i} \prod_{k=1}^K \left\{ \exp \left(-G_k \left[\int_0^{L_{ijk}} \exp \{ \beta^\top X_{ijk}(s) + b_i^\top Z_{ijk}(s) \} d\Lambda_k(s) \right] \right) \right. \\ \left. - \exp \left(-G_k \left[\int_0^{R_{ijk}} \exp \{ \beta^\top X_{ijk}(s) + b_i^\top Z_{ijk}(s) \} d\Lambda_k(s) \right] \right) \right\} \\ \times (2\pi)^{-d_i/2} |\Sigma_i(\gamma)|^{-1/2} \exp \left\{ -\frac{b_i^\top \Sigma_i(\gamma)^{-1} b_i}{2} \right\} db_i$$

NPMLE: treat each Λ_k as a step function

- $t_{k1} < t_{k2} < \dots < t_{km_k}$: distinct values of all $L_{ijk} > 0$ and $R_{ijk} < \infty$ ($i = 1, \dots, n; j = 1, \dots, J_i$)
- λ_{kl} : jump size of Λ_k at t_{kl} ($l = 1, \dots, m_k$)
- $X_{ijk l} = X_{ijk}(t_{kl})$ and $Z_{ijk l} = Z_{ijk}(t_{kl})$

Likelihood (cont.)

Consider the class of frailty-induced transformations:

$$G_k(x) = -\log \int_0^\infty e^{-x\xi} f_k(\xi) d\xi$$

The likelihood can then be written as

$$\begin{aligned} \tilde{L}_n(\theta, \mathcal{A}) = & \prod_{i=1}^n \int_{b_i} \prod_{j=1}^{J_i} \prod_{k=1}^K \int_{\xi_{ijk}} \left[\exp \left\{ -\xi_{ijk} \sum_{t_{kl} \leq L_{ijk}} \exp \left(\beta^\top X_{ijkl} + b_i^\top Z_{ijkl} \right) \lambda_{kl} \right\} \right. \\ & \left. - I(R_{ijk} < \infty) \exp \left\{ -\xi_{ijk} \sum_{t_{kl} \leq R_{ijk}} \exp \left(\beta^\top X_{ijkl} + b_i^\top Z_{ijkl} \right) \lambda_{kl} \right\} \right] f_k(\xi_{ijk}) d\xi_{ijk} \\ & \times (2\pi)^{-d_i/2} |\Sigma_i(\gamma)|^{-1/2} \exp \left\{ -\frac{b_i^\top \Sigma_i(\gamma)^{-1} b_i}{2} \right\} db_i \end{aligned}$$

Poisson data augmentation

Latent variables: for $i = 1, \dots, n; j = 1, \dots, J_i; k = 1, \dots, K; l = 1, \dots, m_k$,

$$W_{ijkl} \stackrel{\text{ind}}{\sim} \text{Poisson} \left\{ \lambda_{kl} \xi_{ijk} \exp \left(\beta^T X_{ijkl} + b_i^T Z_{ijkl} \right) \right\}$$

Equivalent likelihood: Conditional on b_i and ξ_{ijk} , the probability of the event

$$\tilde{\mathcal{O}}_{ijk} = \left(\sum_{t_{kl} \leq L_{ijk}} W_{ijkl} = 0 \right) \cap \left(\sum_{L_{ijk} < t_{kl} \leq R_{ijk}} W_{ijkl} > 0 \right)^{I(R_{ijk} < \infty)}$$

is equal to

$$p(\mathcal{O}_{ijk} | b_i, \xi_{ijk}) = \exp \left\{ -\xi_{ijk} \sum_{t_{kl} \leq L_{ijk}} \exp \left(\beta^T X_{ijkl} + b_i^T Z_{ijkl} \right) \lambda_{kl} \right\} \\ - I(R_{ijk} < \infty) \exp \left\{ -\xi_{ijk} \sum_{t_{kl} \leq R_{ijk}} \exp \left(\beta^T X_{ijkl} + b_i^T Z_{ijkl} \right) \lambda_{kl} \right\}$$

EM algorithm

- Therefore, maximizing $\tilde{L}_n(\theta, \mathcal{A})$ is equivalent to maximizing the likelihood arising from $\{\tilde{\mathcal{O}}_{ijk} : i = 1, \dots, n; j = 1, \dots, J_i; k = 1, \dots, K\}$.
- The maximization can be done through an EM algorithm, treating b_i , ξ_{ijk} and W_{ijkl} as missing data.
- Define $R_{ijk}^* = L_{ijk}I(R_{ijk} = \infty) + R_{ijk}I(R_{ijk} < \infty)$. The complete-data log-likelihood is

$$\sum_{i=1}^n \left\{ \sum_{j=1}^{J_i} \sum_{k=1}^K \left(\sum_{l=1}^{m_k} I(t_{kl} \leq R_{ijk}^*) [W_{ijkl} \log \{ \lambda_{kl} \xi_{ijk} \exp(\beta^T X_{ijkl} + b_i^T Z_{ijkl}) \} \right. \right. \\ \left. \left. - \lambda_{kl} \xi_{ijk} \exp(\beta^T X_{ijkl} + b_i^T Z_{ijkl}) - \log(W_{ijkl}!) \right] + \log f_k(\xi_{ijk}) \right) \\ \left. - \frac{d_i}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma_i(\gamma)| - \frac{b_i^T \Sigma_i(\gamma)^{-1} b_i}{2} \right\}$$

E-step

- In the E-step, we evaluate $\hat{E}(W_{ijkl})$ and some $\hat{E}\{H(\xi_{ijk}, b_i)\}$.
- We use the fact that the joint posterior density of ξ_{ijk} ($j = 1, \dots, J_i$ and $k = 1, \dots, K$) and b_i is proportional to

$$\prod_{j=1}^{J_i} \prod_{k=1}^K p(\mathcal{O}_{ijk} | b_i, \xi_{ijk}) f_k(\xi_{ijk}) \phi(b_i; \Sigma_i(\gamma))$$

- In addition, $E(W_{ijkl} | b_i, \xi_{ijk})$ is given by

$$\frac{I(L_{ijk} < t_{kl} \leq R_{ijk} < \infty) \lambda_{kl} \xi_{ijk} \exp(\beta^T X_{ijkl} + b_i^T Z_{ijkl})}{1 - \exp\left\{-\sum_{L_{ijk} < t_{kl'} \leq R_{ijk}} \lambda_{kl'} \xi_{ijk} \exp(\beta^T X_{ijkl'} + b_i^T Z_{ijkl'})\right\}}.$$

- Then we integrate the above expressions over b_i and ξ_{ijk} using Gaussian quadrature approximations.

M-step

- In the M-step, we first update λ_{kl} ($k = 1, \dots, K$ and $l = 1, \dots, m_k$) by

$$\lambda_{kl} = \frac{\sum_{i=1}^n \sum_{j=1}^{J_i} I(t_{kl} \leq R_{ijk}^*) \hat{E}(W_{ijkl})}{\sum_{i=1}^n \sum_{j=1}^{J_i} I(t_{kl} \leq R_{ijk}^*) \hat{E}\{\xi_{ijk} \exp(\beta^\top X_{ijkl} + b_i^\top Z_{ijkl})\}}$$

- Then we solve the following score equation for β using the one-step Newton-Raphson method:

$$0 = \sum_{i=1}^n \sum_{j=1}^{J_i} \sum_{k=1}^K \sum_{l=1}^{m_k} I(t_{kl} \leq R_{ijk}^*) \hat{E}(W_{ijkl}) \times \left[X_{ijkl} - \frac{\sum_{i'=1}^n \sum_{j'=1}^{J_{i'}} I(t_{kl} \leq R_{i'j'k}^*) X_{i'j'kl} \hat{E}\{\xi_{i'j'k} \exp(\beta^\top X_{i'j'kl} + b_{i'}^\top Z_{i'j'kl})\}}{\sum_{i'=1}^n \sum_{j'=1}^{J_{i'}} I(t_{kl} \leq R_{i'j'k}^*) \hat{E}\{\xi_{i'j'k} \exp(\beta^\top X_{i'j'kl} + b_{i'}^\top Z_{i'j'kl})\}} \right]$$

- Finally, we maximize $-\log |\Sigma_i(\gamma)| - \hat{E}\{b_i^\top \Sigma_i^{-1}(\gamma) b_i\}$ to update γ .

Remarks

- The starting values of the EM algorithm can be $\beta = 0$, $\lambda_{kl} = 1/m_k$, and $\Sigma_i = I_{d_i}$.
- Due to the presence of the random effects, the conditional expectations in the E-step are more complicated than those in Zeng et al. (2016).
- The high-dimensional parameters λ_{kl} are calculated explicitly in the M-step.
- Each iteration of the EM algorithm guarantees an increase in the likelihood.

Variance estimation

- Define the profile likelihood

$$\text{pl}_n(\theta) = \max_{\mathcal{A}} \log L_n(\theta, \mathcal{A}),$$

which can be computed using the same EM algorithm but with fixed θ .

- The covariance matrix of $\hat{\theta}$ can be estimated by

$$\hat{V} = \left(\left[\sum_{i=1}^n \frac{\left\{ \text{pl}_{ni}(\hat{\theta} + h_n \mathbf{e}_j) - \text{pl}_{ni}(\hat{\theta}) \right\} \left\{ \text{pl}_{ni}(\hat{\theta} + h_n \mathbf{e}_k) - \text{pl}_{ni}(\hat{\theta}) \right\}}{h_n^2} \right]_{(j,k)} \right)^{-1},$$

where pl_{ni} denotes the i th cluster's contribution to pl_n , $\mathbf{e}_j$ is the j th canonical vector, and h_n is a constant of order $n^{-1/2}$.

- Compared to the variance estimator in Zeng et al. (2016), this $\hat{V}$ is guaranteed to be positive semidefinite and is more robust w.r.t. the choice of h_n .

Asymptotic theory

Consistency: $\|\hat{\theta} - \theta_0\| + \sum_{k=1}^K \sup_{t \in [0, \tau_k]} |\hat{\Lambda}_k(t) - \Lambda_{0k}(t)| \xrightarrow{a.s.} 0$

Asymptotic normality & semiparametric efficiency:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \tilde{\mathcal{I}}_0^{-1}),$$

where $\tilde{\mathcal{I}}_0$ is the efficient information matrix of θ .

Consistency of variance estimator: $\|n\hat{V} - \tilde{\mathcal{I}}_0^{-1}\|_2 = o_p(1)$.

Convergence rate for $\hat{\Lambda}_k$:

$$E \left[\sum_{j=1}^{J_i} \sum_{k=1}^K \sum_{l=0}^{M_{ijk}} \left\{ \hat{\Lambda}_k(U_{ijkl}) - \Lambda_{0k}(U_{ijkl}) \right\}^2 \right] = O_p \left(n^{-2/3} + \|\hat{\beta} - \beta_0\|^2 + \|\hat{\gamma} - \gamma_0\|^2 \right).$$

Software

The methods developed in Zeng et al. (2016, 2017) have been implemented in **IntCens** (<https://dlin.web.unc.edu/software/intcens>).

Table of Contents

- 1 Chapter 2: Semiparametric regression analysis of interval-censored data
 - Transformation models for interval-censored data
 - Transformation models for multivariate interval-censored data
 - **Marginal models for multivariate interval-censored data**
 - Semiparametric regression models for panel count data

Reference



Xu, Y., Zeng, D., & Lin, D. Y. (2023). Marginal proportional hazards models for multivariate interval-censored data. *Biometrika*, 110(3), 815-830.

Motivation

Random-effects models have several limitations.

- Random effects may not adequately capture the dependence
- Model misspecification may lead to invalid statistical inference
- Computationally demanding
- Interpretation of β does not pertain to population-average effects

Marginal models formulate marginal distributions of multivariate event times through univariate regression models while leaving the dependence structures completely unspecified.

- More robust inference
- Faster and more stable computation
- Interpretation of population-average effects for β

Marginal models

Notation:

- n : number of independent clusters
- J_i : number of subjects in the i th cluster
- K : number of event types
- T_{ijk} : k th event time for the j th subject of the i th cluster
- $X_{ijk}(t)$: potentially time-dependent covariates
- $\lambda_{ijk}(t|X_{ijk})$: marginal hazard function for T_{ijk} conditional on X_{ijk}

Marginal Cox model:

$$\lambda_{ijk}(t|X_{ijk}) = \lambda_k(t) \exp \{ \beta_k^T X_{ijk}(t) \}$$

- β_k : type-specific regression parameters
- $\lambda_k(\cdot)$: arbitrary baseline hazard function
- $\Lambda_k(t) = \int_0^t \lambda_k(s) ds$

Remarks

Marginal Cox model:

$$\lambda_{ijk}(t|X_{ijk}) = \lambda_k(t) \exp \{ \beta_k^T X_{ijk}(t) \}$$

- The dependence structures of the event times within a cluster and between the K types of events are unspecified.
- By letting X_{ijk} depend on k , we allow different sets of covariates for different event types.

Estimation

Data: $\{(L_{ijk}, R_{ijk}, X_{ijk}) : i = 1, \dots, n; j = 1, \dots, J_i; k = 1, \dots, K\}$

Independence working assumption: all event times are independent conditional on covariates

Pseudo-likelihood:

$$\tilde{L}_k(\beta_k, \Lambda_k) = \prod_{i=1}^n \prod_{j=1}^{J_i} \left(\exp \left[- \int_0^{L_{ijk}} \exp \{ \beta_k^\top X_{ijk}(t) \} d\Lambda_k(t) \right] \right. \\ \left. - \exp \left[- \int_0^{R_{ijk}} \exp \{ \beta_k^\top X_{ijk}(t) \} d\Lambda_k(t) \right] \right)$$

Nonparametric maximum pseudo-likelihood estimation:

- Extension of NPMLE
- (β_k, Λ_k) can be estimated using the approach developed in Zeng et al. (2016), i.e., Poisson data augmentation + EM algorithm.

Variance estimation

- Define the **profile pseudo-log-likelihood** for β_k as

$$\text{pl}_k(\beta_k) = \max_{\Lambda_k} \tilde{L}_k(\beta_k, \Lambda_k)$$

- We estimate $\text{Cov}(\hat{\beta}_k, \hat{\beta}_l)$ by the **sandwich covariance estimator**

$$\hat{V}_{kl} = \left\{ D_{h_n}^2 \text{pl}_k(\hat{\beta}_k) \right\}^{-1} \left\{ \sum_{i=1}^n D_{h_n} \text{pl}_{ki}(\hat{\beta}_k) D_{h_n} \text{pl}_{li}(\hat{\beta}_l)^T \right\} \left\{ D_{h_n}^2 \text{pl}_l(\hat{\beta}_l) \right\}^{-1}$$

- ▶ pl_{ki} : i th cluster's contribution to pl_k
- ▶ D_{h_n} and $D_{h_n}^2$: first- and second-order numerical derivatives with perturbation constant $h_n = O(n^{-1/2})$
- ▶ account for the dependence within clusters and between event types

Asymptotic properties

Let $\theta = (\beta_1^\top, \dots, \beta_K^\top)^\top$.

Consistency: $\|\hat{\theta} - \theta_0\| + \sum_{k=1}^K \sup_{t \in [0, \tau_k]} |\hat{\Lambda}_k(t) - \Lambda_{0k}(t)| \xrightarrow{a.s.} 0$

Asymptotic normality: $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \Omega)$

Consistency of variance estimator: $\{n\hat{V}_{kl}\}_{(k,l)}$ is consistent for Ω , **regardless of the dependence structures.**

Simultaneous Inference

- Parameters of interest: $\eta_k = \beta_{k1}$ ($k = 1, \dots, K$)
- Estimators: $\hat{\eta}_k = \hat{\beta}_{k1}$ ($k = 1, \dots, K$)
- Covariance estimator: $\hat{\Psi} = \{\hat{V}_{kl,11}\}_{(k,l)}$
- Global (Wald) test $H_0 : \eta_1 = \dots = \eta_K = 0$

$$W = (\hat{\eta}_1, \dots, \hat{\eta}_K) \hat{\Psi}^{-1} (\hat{\eta}_1, \dots, \hat{\eta}_K)^T \xrightarrow{d} \chi_K^2 \quad \text{under } H_0$$

- To make inference on an overall covariate effect, we can estimate a common parameter $\eta_1 = \dots = \eta_K = \eta$ by

$$\hat{\eta} = \sum_{k=1}^K c_k \hat{\eta}_k$$

- ▶ optimal weights c_k are chosen to minimize $\text{Var}(\hat{\eta})$
- ▶ more efficient than the separate estimators $\hat{\eta}_k$
- ▶ powerful test for no covariate effect on the K events

Table of Contents

- ① Chapter 2: Semiparametric regression analysis of interval-censored data
 - Transformation models for interval-censored data
 - Transformation models for multivariate interval-censored data
 - Marginal models for multivariate interval-censored data
 - Semiparametric regression models for panel count data

Reference

-  Zeng, D., & Lin, D. Y. (2021). Maximum likelihood estimation for semiparametric regression models with panel count data. *Biometrika*, 108(4), 947-963.

Panel count data

- **Panel count data** arise when only the number of recurrent events between successive examinations can be observed.
 - ▶ number of tumors in a cancer patient
 - ▶ number of damaged joints in a psoriatic arthritis patient
 - ▶ number of decayed teeth in a child
- Investigators are often interested in evaluating the effects of covariates (e.g., treatment) on the recurrent event process.
- Challenges:
 - ▶ Unknown recurrent event times
 - ▶ Within-subject correlations of recurrent event times
 - ▶ Within-subject correlations between different types of recurrent events

Notation

- n : number of subjects
- K : number of types of recurrent events
- $N_{ik}(t)$: number of the k th type of event that the i th subject has experienced by time t ($i = 1, \dots, n$ and $k = 1, \dots, K$); non-homogeneous Poisson process
- $X_i(t)$: potentially time-dependent covariates
- b_{ik} : random effects for the k th type of event
- ξ_i : random effects shared by the K types of events
- $\lambda_{ik}(t|X_i, b_{ik}, \xi_i)$: conditional intensity function for $N_{ik}(t)$

Proportional intensity models

$$\lambda_{ik}(t|X_i, b_{ik}, \xi_i) = \lambda_k(t) \exp \left\{ \beta_k^T X_i(t) + b_{ik}^T Z_i(t) + \xi_i^T \tilde{Z}_i(t) \right\}$$

- $\lambda_k(t)$: unknown baseline intensity function
- β_k : unknown type-specific regression parameters
- $Z_i(t)$ and $\tilde{Z}_i(t)$: contain 1 and part of $X_i(t)$
- $b_{ik} \sim N(0, \Sigma_k)$: accounts for within-subject correlations among recurrent event times of the k th type
- $\xi_i \sim N(0, \Psi)$: accounts for within-subject correlations between different recurrent event processes
- b_{ik} ($k = 1, \dots, K$) and ξ_i are mutually independent.
- If $K = 1$, ξ_i is omitted.

Data

Panel count data for the i th subject:

- Examination times:

$$U_{ik} = (0 = U_{ik0}, U_{ik1}, \dots, U_{ikM_{ik}}), \quad \text{for } k = 1, \dots, K$$

- Event counts:

$$\Delta_{ik} = (\Delta_{ik1}, \dots, \Delta_{ikM_{ik}}), \quad \text{for } k = 1, \dots, K$$

with $\Delta_{ikj} = N_{ik}(U_{ikj}) - N_{ik}(U_{ik,j-1})$.

- Covariates: $X_i(t)$

Independent censoring assumption: $(U_{i1}, \dots, U_{iK})$ are independent of $(N_{i1}, \dots, N_{iK})$, $(b_{i1}, \dots, b_{iK})$, and ξ_i conditional on $X_i(\cdot)$.

Likelihood

- $\{\Delta_{ikj}\}_{j=1}^{M_{ik}} \stackrel{\text{ind}}{\sim} \text{Poisson with means } \int_{U_{ik,j-1}}^{U_{ikj}} \lambda_{ik}(t|X_i, b_{ik}, \xi_i) dt$
- The likelihood is proportional to

$$\prod_{i=1}^n \left[\int_{\xi_i} \phi(\xi_i; \Psi) \prod_{k=1}^K \int_{b_{ik}} \phi(b_{ik}; \Sigma_k) \prod_{j=1}^{M_{ik}} \frac{\left\{ \int_{U_{ik,j-1}}^{U_{ikj}} e^{\beta_k^T X_i(t) + b_{ik}^T Z_i(t) + \xi_i^T \tilde{Z}_i(t)} d\Lambda_k(t) \right\}^{\Delta_{ikj}}}{\Delta_{ikj}!} \right. \\ \left. \times \exp \left\{ - \int_0^{U_{ikM_{ik}}} e^{\beta_k^T X_i(t) + b_{ik}^T Z_i(t) + \xi_i^T \tilde{Z}_i(t)} d\Lambda_k(t) \right\} db_{ik} d\xi_i \right]$$

- ▶ $\Lambda_k(t) = \int_0^t \lambda_k(s) ds$: cumulative baseline intensity function
- ▶ $\phi(\cdot; \Sigma)$: multivariate normal density with mean 0 and covariance matrix Σ

Estimation

NPMLE:

- $0 < t_{k1} < t_{k2} < \dots < t_{km_k}$: unique values of U_{ik} ($i = 1, \dots, n$)
- λ_{kl} : jump size of Λ_k at t_{kl} ($l = 1, \dots, m_k$)
- $X_{ikl} = X_i(t_{kl})$, $Z_{ikl} = Z_i(t_{kl})$, and $\tilde{Z}_{ikl} = \tilde{Z}_i(t_{kl})$

New likelihood:

$$\prod_{i=1}^n \left\{ \int_{\xi_i} \phi(\xi_i; \Psi) \prod_{k=1}^K \int_{b_{ik}} \phi(b_{ik}; \Sigma_k) \prod_{j=1}^{M_{ik}} \frac{\left(\sum_{l: t_{kl} \in (U_{ik,j-1}, U_{ikj}]} \lambda_{kl} e^{\beta_k^T X_{ikl} + b_{ik}^T Z_{ikl} + \xi_i^T \tilde{Z}_{ikl}} \right)^{\Delta_{ikj}}}{\Delta_{ikj}!} \right. \\ \left. \times \exp \left(- \sum_{l: t_{kl} \leq U_{ikM_{ik}}} \lambda_{kl} e^{\beta_k^T X_{ikl} + b_{ik}^T Z_{ikl} + \xi_i^T \tilde{Z}_{ikl}} \right) db_{ik} d\xi_i \right\}$$

Direct maximization is infeasible due to lack of analytic expressions for λ_{kl} .

Poissonization

- We introduce independent latent Poisson variables W_{ikl} with means $\lambda_{kl} e^{\beta_k^T \mathbf{x}_{ikl} + b_{ik}^T \mathbf{z}_{ikl} + \xi_i^T \tilde{\mathbf{z}}_{ikl}}$, for $i = 1, \dots, n$; $k = 1, \dots, K$; $l = 1, \dots, m_k$.
- It is easy to see that the likelihood for Δ_{ikj} is the same as the likelihood for $\sum_{l: \mathbf{t}_{kl} \in (U_{ik,j-1}, U_{ikj}]} W_{ikl} = \Delta_{ikj}$.
- Thus, we can maximize the likelihood through an EM algorithm, with W_{ikl} , b_{ik} , and ξ_i as missing data.

EM algorithm

The complete-data log-likelihood is

$$\begin{aligned} \sum_{i=1}^n & \left[-\frac{1}{2} \log(2\pi)^q |\Psi| - \frac{1}{2} \xi_i^\top \Psi^{-1} \xi_i + \sum_{k=1}^K \left\{ -\frac{1}{2} \log(2\pi)^p |\Sigma_k| - \frac{1}{2} b_{ik}^\top \Sigma_k^{-1} b_{ik} \right\} \right. \\ & + \sum_{k=1}^K \sum_{l=1}^{m_k} I(t_{kl} \leq U_{ikM_{ik}}) \left\{ W_{ikl} \left(\log \lambda_{kl} + \beta_k^\top X_{ikl} + b_{ik}^\top Z_{ikl} + \xi_i^\top \tilde{Z}_{ikl} \right) \right. \\ & \left. \left. - \lambda_{kl} e^{\beta_k^\top X_{ikl} + b_{ik}^\top Z_{ikl} + \xi_i^\top \tilde{Z}_{ikl}} - \log W_{ikl}! \right\} \right]. \end{aligned}$$

E-step

- In the E-step, we compute $\hat{E}(W_{ikl})$ and some $\hat{E}\{H(b_{ik}, \xi_i)\}$.
- For any $t_{kl} \in (U_{ik,j-1}, U_{ikj}]$, conditional on Δ_{ikj} , the covariates and random effects, W_{ikl} follows a binomial distribution with success probability

$$p_{ikl} = \frac{\lambda_{kl} e^{\beta_k^T X_{ikl} + b_{ik}^T Z_{ikl} + \xi_i^T \tilde{Z}_{ikl}}}{\sum_{q: t_{kq} \in (U_{ik,j-1}, U_{ikj}]} \lambda_{kq} e^{\beta_k^T X_{ikq} + b_{ik}^T Z_{ikq} + \xi_i^T \tilde{Z}_{ikq}}}.$$

Thus, $\hat{E}(W_{ikl}) = \Delta_{ikj} \hat{E}(p_{ikl})$.

E-step (cont.)

- In addition, the joint posterior density of $\{b_{ik}\}_{k=1}^K$ and ξ_i is proportional to

$$\phi(\xi_i; \Psi) \prod_{k=1}^K \left\{ \phi(b_{ik}; \Sigma_k) \prod_{j=1}^{M_{ik}} \frac{\left(\sum_{l: \mathbf{t}_{kl} \in (U_{ik,j-1}, U_{ikj}] } \lambda_{kl} e^{\beta_k^\top \mathbf{x}_{ikl} + b_{ik}^\top \mathbf{z}_{ikl} + \xi_i^\top \tilde{\mathbf{z}}_{ikl}} \right)^{\Delta_{ikj}}}{\Delta_{ikj}!} \right. \\ \left. \times \exp \left(- \sum_{l: \mathbf{t}_{kl} \leq U_{ikM_{ik}}} \lambda_{kl} e^{\beta_k^\top \mathbf{x}_{ikl} + b_{ik}^\top \mathbf{z}_{ikl} + \xi_i^\top \tilde{\mathbf{z}}_{ikl}} \right) \right\}$$

- The conditional expectations can then be calculated, with integrals over b_{ik} and ξ_i approximated by Gauss–Hermite quadrature.

M-step

- In the M-step, we first update λ_{kl} by

$$\lambda_{kl} = \frac{\sum_{i=1}^n I(U_{ikM_{ik}} \leq t_{kl}) \hat{E}(W_{ikl})}{\sum_{i=1}^n I(U_{ikM_{ik}} \leq t_{kl}) \hat{E}(e^{\beta_k^T X_{ikl} + b_{ik}^T Z_{ikl} + \xi_i^T \tilde{Z}_{ikl}})}$$

- Then we update β_k by applying the one-step Newton-Raphson method to the score equation

$$0 = \sum_{i=1}^n \sum_{l=1}^{m_k} I(U_{ikM_{ik}} \leq t_{kl}) \hat{E}(W_{ikl}) \left\{ X_{ikl} - \frac{\sum_{i'=1}^n I(U_{i'kM_{i'k}} \leq t_{kl}) \hat{E}(e^{\beta_k^T X_{i'kl} + b_{i'k}^T Z_{i'kl} + \xi_{i'}^T \tilde{Z}_{i'kl}}) X_{i'kl}}{\sum_{i'=1}^n I(U_{i'kM_{i'k}} \leq t_{kl}) \hat{E}(e^{\beta_k^T X_{i'kl} + b_{i'k}^T Z_{i'kl} + \xi_{i'}^T \tilde{Z}_{i'kl}})} \right\}$$

- Finally, we set $\Sigma_k = n^{-1} \sum_{i=1}^n \hat{E}(b_{ik}^{\otimes 2})$ and $\Psi = n^{-1} \sum_{i=1}^n \hat{E}(\xi_i^{\otimes 2})$.

Prediction

- We can use the event history to improve the prediction of future events.
- The event history at the current examination time t_0 consists of

$$\mathcal{H}(t_0) = \{n_k = N_k(t_0) : k = 1, \dots, K\}$$

- The key is to update the posterior density of the random effect ξ based on $\mathcal{H}(t_0)$, which is proportional to

$$\begin{aligned} \tilde{\phi}(\xi; \mathcal{H}(t_0)) &= \phi(\xi; \hat{\Psi}) \prod_{k=1}^K \int_{b_k} \left\{ \int_0^{t_0} e^{\hat{\beta}_k^T X(t) + b_k^T Z(t) + \xi^T \tilde{Z}(t)} d\hat{\Lambda}_k(t) \right\}^{n_k} \\ &\quad \times \exp \left\{ - \int_0^{t_0} e^{\hat{\beta}_k^T X(t) + b_k^T Z(t) + \xi^T \tilde{Z}(t)} d\hat{\Lambda}_k(t) \right\} \phi(b_k; \hat{\Sigma}_k) db_k \end{aligned}$$

- Then, the new event count of the k th type at $t_1 > t_0$ can be predicted by

$$\frac{\int_{\xi} \tilde{\phi}(\xi; \mathcal{H}(t_0)) \int_{t_0}^{t_1} e^{\hat{\beta}_k^T X(t) + b_k^T Z(t) + \xi^T \tilde{Z}(t)} d\hat{\Lambda}_k(t) d\xi}{\int_{\xi} \tilde{\phi}(\xi; \mathcal{H}(t_0)) d\xi}$$

Asymptotic properties

Let θ contain β_k and the upper triangular elements of Σ_k and Ψ ($k = 1, \dots, K$).

Consistency: $\|\hat{\theta} - \theta_0\| + \sum_{k=1}^K \sup_{t \in [0, \tau_k]} |\hat{\Lambda}_k(t) - \Lambda_{0k}(t)| \xrightarrow{a.s.} 0$

Asymptotic normality & semiparametric efficiency: $\sqrt{n}(\hat{\theta} - \theta_0)$ converges weakly to a mean-zero normal random vector whose covariance matrix attains the semiparametric efficiency bound.

Variance estimation: The limiting covariance matrix of $\sqrt{n}(\hat{\theta} - \theta_0)$ can be consistently estimated by the inverse of the matrix whose (j, l) th element is

$$n^{-1} \sum_{i=1}^n \left\{ \frac{\text{pl}_{ni}(\hat{\theta} + h_n e_j) - \text{pl}_{ni}(\hat{\theta})}{h_n} \right\} \left\{ \frac{\text{pl}_{ni}(\hat{\theta} + h_n e_l) - \text{pl}_{ni}(\hat{\theta})}{h_n} \right\}$$

- $\text{pl}_{ni}(\theta)$: i th subject's contribution to the profile likelihood for θ
- $h_n = O(n^{-1/2})$

Concluding remarks

- The rationale behind Poisson data augmentation is that conditional on latent variables, the counting process $N(t)$ is a non-homogeneous Poisson process with intensity function the same as the hazard/intensity function for the failure time.
- With interval-censored data, the convergence rate of $\hat{\Lambda}$ is usually slower than $\sqrt{n}$. In all the papers discussed, $\hat{\Lambda}$ converges at a $n^{1/3}$ rate.
- However, the finite-dimensional component of the estimators is still asymptotically normal and efficient, and the limiting variance can be consistently estimated using profile likelihood.

Related work

-  Mao, L., Lin, D. Y., & Zeng, D. (2017). Semiparametric regression analysis of interval-censored competing risks data. *Biometrics*, 73(3), 857-865.
-  Gao, F., Zeng, D., & Lin, D. Y. (2018). Semiparametric regression analysis of interval-censored data with informative dropout. *Biometrics*, 74(4), 1213-1222.
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-  Xu, Y., Zeng, D., & Lin, D. Y. (2024). Proportional rates models for multivariate panel count data. *Biometrics*, 80(1), ujad011.
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